

LA HW

Included exercises: 1, 5, 13, 17, 19, 25, 29, 33, 37

Exercises 1–12: Range

Find a generating set for the range of each linear transformation T .

Original Exercise 1

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 + 3x_2 \\ 4x_1 + 5x_2 \end{bmatrix}.$$

Original Exercise 5

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3, \quad T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 + x_2 + x_3 \\ 2x_1 + 2x_2 + 3x_3 \\ 4x_1 + x_2 \end{bmatrix}.$$

Exercises 13–23: Null Space and One-to-One

Find a generating set for the null space of T , and use it to determine whether T is one-to-one.

Original Exercise 13

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_2 \\ x_1 + x_2 \end{bmatrix}.$$

Original Exercise 17

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3, \quad T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + 2x_2 + x_3 \\ x_1 + 3x_2 + 2x_3 \\ 2x_1 + 5x_2 + 3x_3 \end{bmatrix}.$$

Original Exercise 19 $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is defined by $T(\mathbf{v}) = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^3$.

Exercises 24–31: Standard Matrix and One-to-One

Find the standard matrix of T , and use it to determine whether T is one-to-one.

Original Exercise 25

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 + 3x_2 \\ 4x_1 + 5x_2 \end{bmatrix}.$$

Original Exercise 29

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3, \quad T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 - x_2 \\ x_2 - x_3 \\ x_1 - x_3 \end{bmatrix}.$$

Exercises 32–40: Standard Matrix and Onto

Find the standard matrix of T , and use it to determine whether T is onto.

Original Exercise 33

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 + 3x_2 \\ 4x_1 + 5x_2 \end{bmatrix}.$$

Original Exercise 37

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3, \quad T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_2 - 2x_3 \\ x_1 - x_3 \\ -x_1 + 2x_2 - 3x_3 \end{bmatrix}.$$